Harmonic cavitation

An efficient cavitation heater: a dimpled metal drum at each outlet, spinning on one shaft in a fixed housing, sized from published rotational cavitation reactors, with the Mandelbulb cavity evolved as a challenger — register entry H46

kronspice simulation report · 2026-11-08 · NSGA-II seed 1, 80 × 80

© 2026 fdamador@comcast.net

Summary

Two dimpled metal drums, one at each outlet, turn on one shaft driven by a motor or a Tesla turbine. Each drum carries rows of hemispherical dimples facing a matching ring of dimples in a fixed housing across a small gap. Water is cavitated in the gap and leaves through the end caps as hot water or steam, to heat homes. The measure is **cavitation per joule**: η = collapse energy, $(p - p_v)\cdot V_{vapour}$ per cavity cycle, divided by shaft work. Heat per joule is no measure, because all shaft work ends as heat whatever the cavity.

- **Heater design (3 kW shaft):** η = 0.189, 565 W of collapse energy from 2998 W. Drums of 84 mm radius with 2 rows of Ø 6 mm hemispheres, 26 per row, against a housing dimple ring; gap 0.8 mm, 1,830 rpm (16 m/s), outlet 1.18 bar, σ = 0.881. With a plain housing the same drums give 0.117.
- **Size is free:** the evolved front is flat in η (0.1886–0.1891) from 1.1 kW to 3.0 kW of shaft power, so the drums can be sized to the heating load.
- **One speed for every dimple:** on a drum every dimple runs at the same speed and cavitation number, so one shaft speed puts them all at their best.
- **The design rules hold** in every rerun with the declared constants changed: full hemispheres, dimples on the housing, the smallest gap, σ ≈ 0.88. The absolute η is less certain: 0.09–0.47, mostly with the declared drag coefficient.
- **Mandelbulb (H46):** evolved over its power n and the machine, the best is η = 0.187 (n = 14). Each n separately gives 0.158–0.186. With its fine detail removed the same bulb gives 0.191: a smooth lobed ring does as well as dimples, and the fractal detail lowers it.
- **Energy:** heat is never more than the electricity. A motor on the shaft gives 0.170 collapse per electrical joule; a pump driving a Tesla turbine gives 0.040, a quarter. A heat pump on the motor's 3.3 kW would deliver 11.7 kW of heat.
- **Closed steam loop (the user's layout, Part 10):** water enters at the drums' outer top, the drums make steam, the steam drives the Tesla turbine on the same shaft, and it is condensed (the home's heat) and pumped back. The turbine returns only 115 W of 3,000 W (Carnot allows 13 % between 111 °C and 60 °C, and the turbine is declared at 30 % of ideal), so a motor supplies the rest and the loop cannot run itself. Boiling in the drums drops η to 0.011: steam that leaves has not collapsed. Pressurized drums (3 bar, no boiling) with a flash tank keep η = 0.156.

© 2026 fdamador@comcast.net

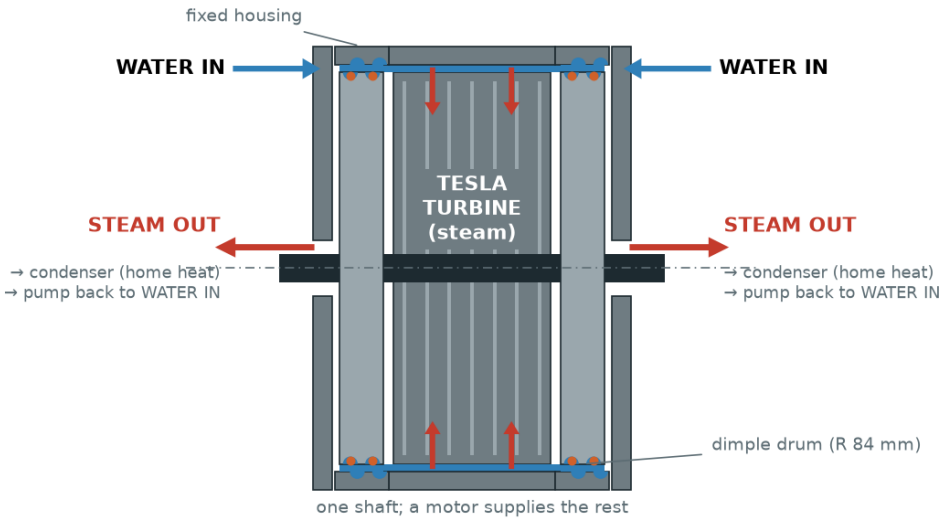


Figure 1. The design in section, in the user's loop. Water enters at each drum's outer top; the drums (orange: cavitation in the dimples, facing the housing's dimple ring across the gap, blue: water, gap drawn ×4) heat it; the Tesla turbine's disks reach the drums' radius, so the same gap runs on over the turbine's rim and the water enters the turbine at its periphery; the steam drives the turbine on the same shaft and leaves along the shaft to the condenser (the home's heat), from which the water is pumped back.

What is being tested

Hypothesis (H46, the user's). Cavities shaped as a Mandelbulb, the 3-D power-n fractal of White and Nylander, give the most cavitation on a rotor turned by a Tesla turbine. Their harmonics and implosions gather in the middle. The heat, as steam, heats homes. The power n is to be evolved. The user then fixed the layout: a dimple drum at each outlet.

Standard prediction. Rotational cavitation reactors are well studied: cavities on a rotor passing a fixed surface or fixed cavities across a gap of a few millimetres. Their measured heaters return 80–90 % of the electricity as heat (Kwon & Yoon 2012; Yoon group). All shaft work becomes heat, sound and steam. Focused collapses raise the peak pressure, not the energy.

The model

A reduced model calibrated on published reactors, not CFD (CLAUDE.md §1 keeps CFD and bubble dynamics out of the engine; the bubble enters as Rayleigh–Plesset). Every constant is either fitted to a published number or declared, and the declared ones are run at the ends of their ranges.

quantity	value	origin
inception σ_i	1.105 (bracket 0.968–1.243)	Xue et al. 2024: none at 3000 rpm, bubbles at 3400 (Ø 80 mm); 1 atm declared
best σ	0.775	Xue 2024: strongest pulsation at 3800 rpm
pulsation St = fW/U	0.446; also 0.28	Xue 2024 (1122 Hz at 3600 rpm); cloud shedding (Stutz & Legoupil)
vapour law	$g = x \cdot e^{1-x}$, $x = (\sigma_i - \sigma)/\Delta$, $\Delta = 0.272$	Δ fitted so Xue's collapse rate peaks at 3800 rpm
vapour at the peak κ_v	0.2 of the cavity	declared, order of Sun et al. 2021's 5.44×10^{-8} m³ beside Ø 10 mm units
cavity drag C_d	0.25 on the projected area	declared, order of Sun 2021's 1.63 N·m (one 1/32 sector assumed)
housing pulse	$0.3 \cdot \frac{1}{2}\rho U^2 \cdot (1.5\ \text{mm/gap})^{0.60}$	exponent fitted to Petkovšek 2013 (0.7 bar at 3.5 mm, 1.7 at 0.8 mm); 0.3 declared
wall friction	max(μU/gap, turbulent flat plate)	Schlichting $c_f = 0.0592\ \text{Re}^{-0.2}$; drum surface and both end faces (hub 10 mm)
gap pressure	the outlet pressure	set at the drum's end (a gene)

Per cavity, the collapse energy per event is $(p_{local} - p_v + \text{housing pulse}) \cdot V_{vapour}$. The event rate is the cloud rate St·U/W on a plain housing, and the other ring's passing rate, capped at the cloud rate, with housing dimples. It is

© 2026 fdamador@comcast.net

never faster than the Rayleigh collapse allows. Each design is scored at the worse St. Collapse energy is a part of the cavity drag, and a design claiming more is rejected.

The shaft as a kronspice circuit. In the mobility analogy (angular speed across, torque through) the shaft is a speed source ω driving one conductance per loss: friction, cavity drag less collapse, and collapse. Each absorbs ω·torque. The GA ranks every design by the power the operating point books into the collapse branch over the power the source gives. The ledger closes to round-off (residual 0).

Part 1–2. Calibration and drum sizing

The resonance rule sets each ring's dimple count so the passing frequency equals the cloud frequency: $N = 2\pi r \cdot St/W$, a pitch of about 2.2 W. On Xue's own rotor it gives 18.7 grooves against the 20 built.

A dimple works only within a window of $\sigma = (p - p_v)/(\tfrac{1}{2}\rho U^2)$. At 1 atm cavitation starts near U = 13.3 m/s and is strongest near 15.9 m/s, so a drum of 50 mm radius cavitates from 2,550 rpm and is strongest at 3,040 rpm; one of 84 mm, from 1,520 and at 1,810 rpm. On a drum every dimple runs at the same speed, so one shaft speed suits them all.

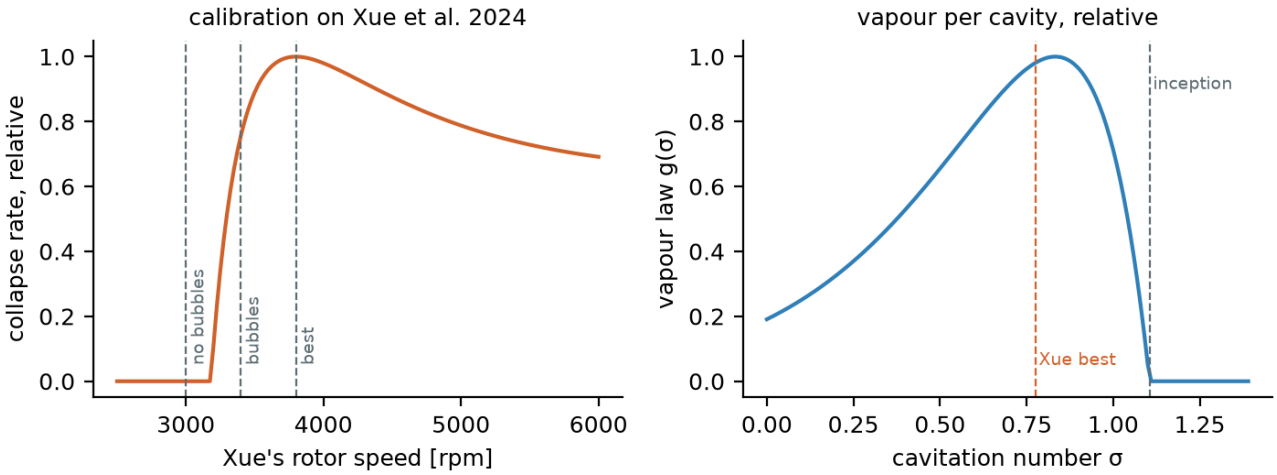


Figure 2. Left: the model's collapse rate on Xue's rotor against speed, with the measured no-bubble, first-bubble and best speeds. Right: the fitted vapour law g(σ), zero above inception and falling again as the cavity chokes.

Part 3–4. The search

NSGA-II (population 80, 80 generations) over drum radius, rows, dimple diameter and depth, rotor and housing pitch, housing dimples on or off, gap, speed and outlet pressure. Two objectives: η and collapse power. Constraints: shaft ≤ 3 kW, land ≥ 1 mm, energy consistency. 6 feasible designs on the front, all within η 0.1886–0.1891; the heater design is the one with the most collapse power within 0.5 % of the best η.

	heater design (two drums)
drum radius / rows	84.3 mm / 2
dimples	Ø 6.0 mm, depth W/2 (hemisphere), 26 per row, pitch 3.3 W
housing	dimple ring, pitch 1.2 W (73 per ring)
gap / speed / outlet	0.80 mm / 1,828 rpm (16.1 m/s) / 1.18 bar
σ	0.881
shaft = friction + drag	2998 W = 76 + 2922 W
collapse energy / η	565 W / 0.1886

© 2026 fdamador@comcast.net

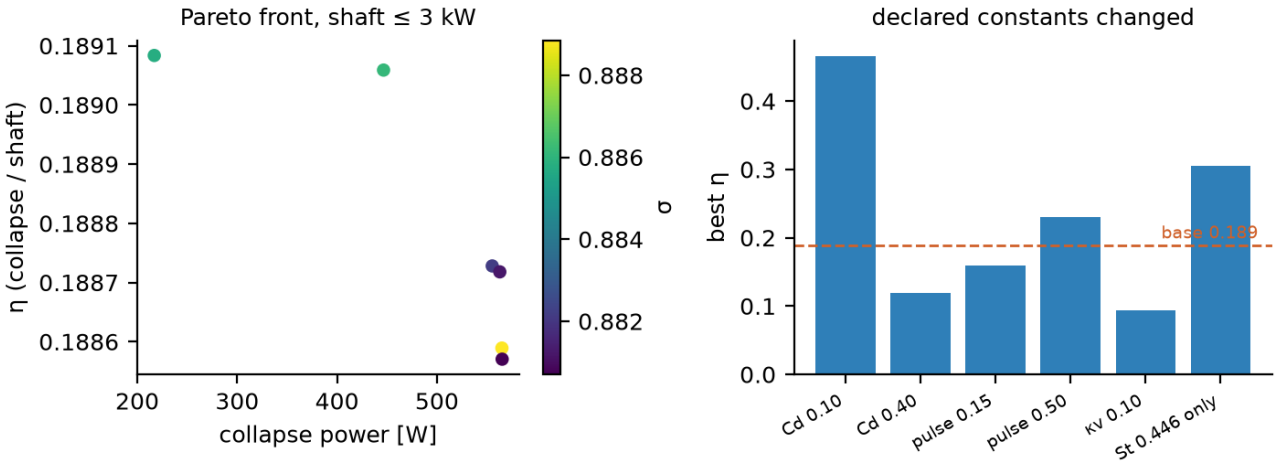


Figure 3. Left: the Pareto front (η against collapse power, colour σ): flat in η, so the size follows the load. Right: the best η when the search is rerun with one declared constant changed. The drag coefficient dominates.

The most collapse per joule comes at σ ≈ 0.88, a slower speed than the most collapse per second (σ = 0.775). Every search drives the gap to the smallest value in the data (0.8 mm), so the true optimum may be smaller still, outside the calibrated range.

Part 5. The gap as an acoustic ring

The annular gap is a kronspice ladder closed on itself: series inertance ρΔx/A, shunt compliance AΔx/ρc², 64 sections, driven by a 1 m³/s volume-velocity source, Q 2000. Its peaks equal the lattice's exact frequencies to 10^{-4}, and these tend to m·c/2πR within the lattice's sin x/x. The 26-dimple pattern passes at 792 Hz, while its own mode 26 is at 73.7 kHz in clear water. The two meet only if the bubbly mixture's sound speed falls to the rotor speed, 16.1 m/s. Wood's minimum at 118 kPa is 21.7 m/s, so this resonance is out of reach at this design.

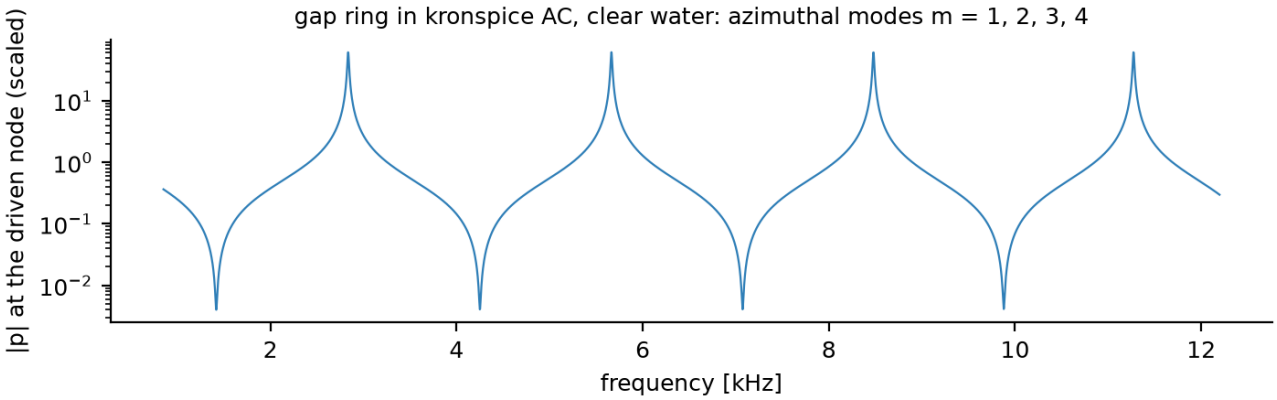


Figure 4. |p| at the driven node of the gap ring (kronspice AC), clear water: the first four azimuthal modes.

Part 6. One collapse, and collapses on the axis

The cavity's vapour (11.1 mm³, a sphere of R_0 = 1.38 mm) collapses under 171 kPa: the local 114 kPa plus the housing pulse of 57 kPa. Empty, it collapses in 96.66 µs against Rayleigh's 96.69 µs. With 10^{-3} gas, surface tension and viscosity, its energy books close to 2.1×10^{-6}.

© 2026 fdamador@comcast.net

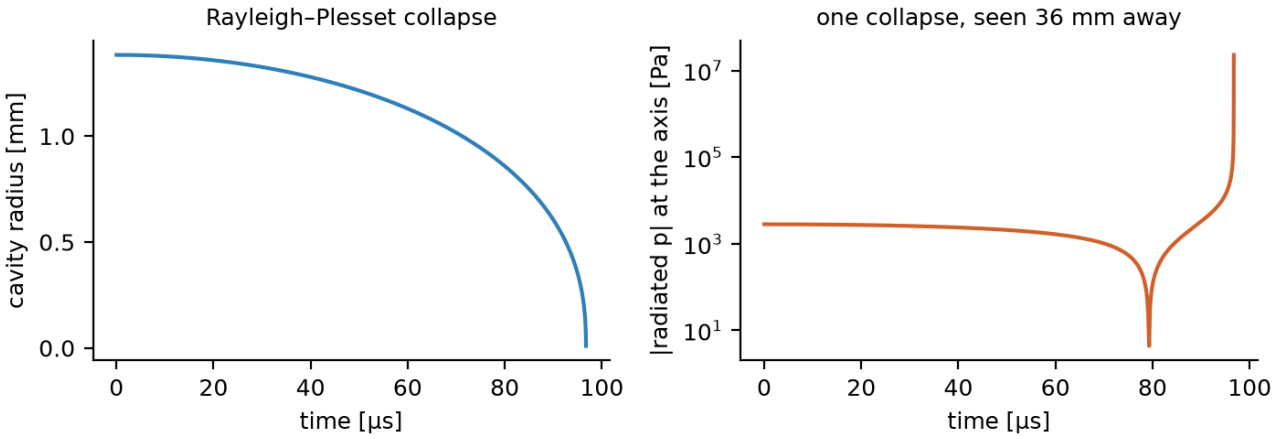


Figure 5. The heater design's cavity: radius, and the pressure it radiates to the axis 84 mm away (incompressible Rayleigh–Plesset; the final peaks are overstated).

Gathering in the middle. Rotor and housing rings line up gcd(N_r, N_h) cavities at a time. With 26 and 73 that is one; with equal counts all 26 collapse together and their pulses reach the axis together, raising the peak there by up to 26×. The energy is still only the 26 collapses' share of the shaft work. These figures are upper bounds (linear superposition of incompressible near fields). In a solid drum the axis is metal, so the focused pulse goes into the rotor; a hollow drum with a water core would put it in the water.

Part 7. Energy books

drive	electric	shaft	heat to water	heat to room	collapse / electric J	steam
motor on the shaft	3331 W	2998 W	2998 W	333 W	0.170	4.13 kg/h
pump + Tesla turbine	14275 W	2998 W	12848 W	1428 W	0.040	17.70 kg/h

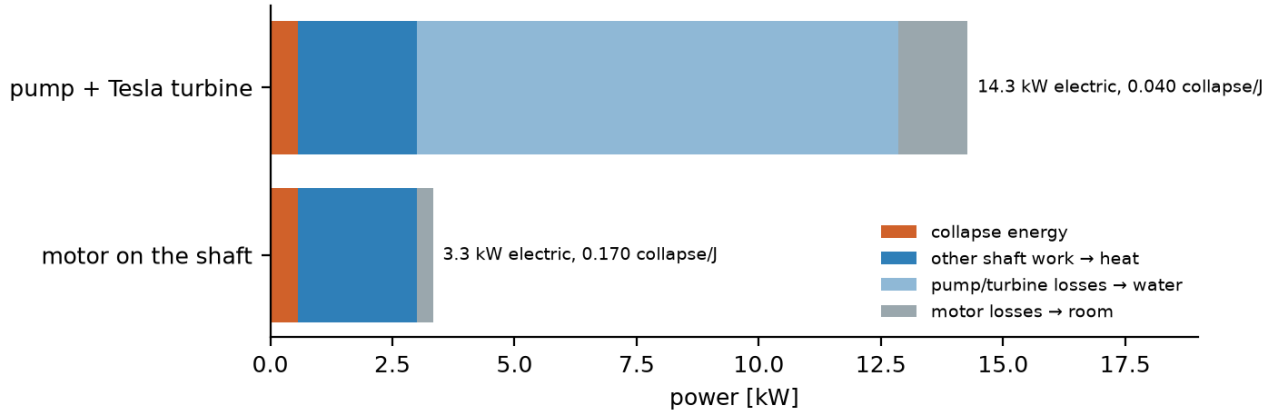


Figure 6. Where the electricity goes. Declared: motor 0.90, pump 0.70, Tesla turbine 0.30 (to be measured).

The turbine's and pump's losses also warm the water, so heat per joule is similar, but cavitation per joule is four times lower. A heat pump (COP 3.5) on the same electricity gives 3.5× the heat of either.

Part 8. The Mandelbulb challenger (H46)

The bulb: $v \to v^n + c$ in spherical form, polar convention ($\theta = \arccos z/r$), 10 iterations, detail below 0.3 mm (metal printing) removed. Tested: both conventions have $(n-1)$-fold symmetry about the axis; in the latitude convention ($\theta = \arcsin z/r$) the equator is exactly the degree-n Multibrot set; n = 8 shows 7 lobes. The bulb's latitude rings, scaled to the drum radius, form the drum's surface, and each valley between lobes is a cavity with its own width, depth and volume.

© 2026 fdamador@comcast.net

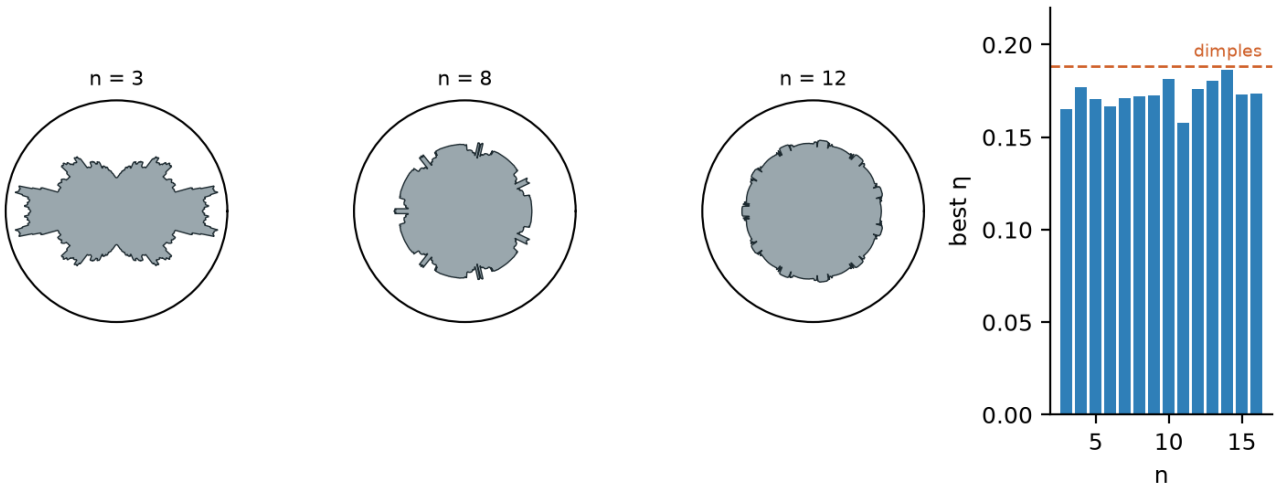


Figure 7. Left: the equator of the n = 3, 8 and 12 bulbs (outer circle: the drum radius). Right: the best η for each n, each from its own NSGA-II search (50 × 50), against the dimple drums.

- NSGA-II over n and the machine (60 × 60): best η = 0.187 at n = 14, 106 cavities, on 31 mm drums at 6,740 rpm.
- The n = 8 bulb put on the heater's 84 mm drums without re-optimizing gives 0.099 at 4.7 kW: the bulb needs its own size and speed.
- Fine detail removed (3 iterations against 10): η = 0.191, as good as the dimples (0.189). The small structure adds drag faster than it adds collapse.
- Status: **bounded** in this model. The Mandelbulb works as a ring of lobes; a smooth lobed ring matches the dimples, and the fractal detail does not help.

Part 10. The closed steam loop

The user's layout: water in at each drum's outer top; the drums' cavitation and shaft work raise steam; the steam drives the Tesla turbine on the drums' own shaft; the exhaust is condensed at the outlet, where its heat goes to the home, and the condensate is pumped back. A motor on the same shaft supplies what the turbine does not. Saturation pressures from IAPWS-IF97 (verified to 10^{-8} at 300 K); steam states from the IAPWS-95 tables; declared: turbine isentropic efficiency 0.30, condenser 0.2 bar (60 °C, warm enough for radiators), motor 0.90, pump 0.70.

Cavitation needs subcooled water. Collapse energy is $(p - p_v)\cdot V$. As the water warms, its vapour pressure p_v rises toward the drum pressure and the collapse fades; at boiling it is gone, and steam that boils off and leaves the drum has not collapsed. So two ways to raise the steam are compared. **A:** the drums are the boiler at 1.5 bar. **B:** the drums run at 3 bar, where water at 100–120 °C is still well below boiling, and the hot water is throttled into a flash tank at 1 atm, whose steam drives the turbine while the remaining hot water returns to the drums.

cycle (3 kW shaft)	steam	turbine	motor shaft	electric	home heat	collapse	η	collapse / elec J
A: boil in the drums, 1.5 bar	4.42 kg/h	115 W	2885 W	3206 W	2885 W	33 W	0.011	0.010
B: subcooled drums 3 bar + flash	4.47 kg/h	93 W	2907 W	3239 W	2914 W	468 W	0.156	0.144

© 2026 fdamador@comcast.net

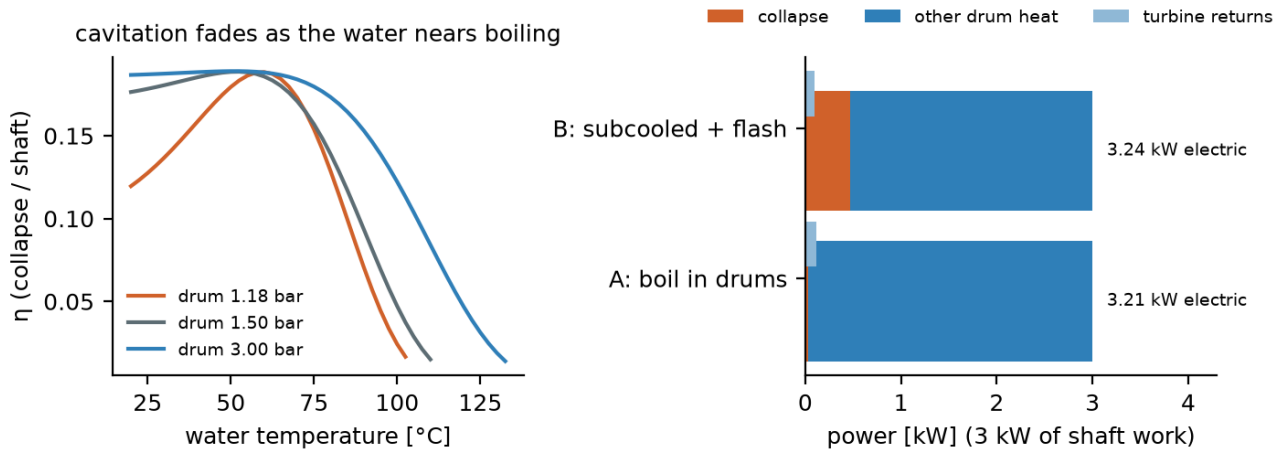


Figure 8. Left: η against water temperature at three drum pressures (heater drums at the speed best for 60 °C water); it falls to zero as the water nears boiling, later at higher pressure. Right: the two loops at 3 kW of shaft work: the collapse share, the rest of the drum heat, and the little the turbine returns.

- **The loop cannot run itself.** Heat becomes work at most at the Carnot efficiency, 13 % between 111 °C and 60 °C, and a Tesla turbine at 30 % of ideal returns 3–4 % of the shaft work. The motor supplies about 96 %.
- **The turbine does not add heat.** The home receives the condenser's heat, about the electricity less the motor's losses, as with no turbine at all. The books close (electricity = home + room to 10^{-12} W), and kronspice's shaft circuit, with the motor as a speed source and the turbine as a torque source, has the motor supply $P_{shaft} - P_{turbine}$ with a closed ledger.
- **Boiling wastes the cavitation:** 91 % of the heat in A goes into boiling, where nothing collapses, so η falls to 0.011. Subcooled drums with a flash tank keep η = 0.156 (the hot water at 3 bar has about the same $p - p_v$ as cold water at 1 atm).
- Collapse in hot water is further reduced by heat transfer into the bubble (the thermal effect, Brennen 1995 §2.3), not modelled here, so the hot-water values are upper bounds.

Limits of this study

- Reduced model, not CFD. Calibrated on one reactor's grooves (Xue 2024) and applied to hemispheres; the drag, vapour and pulse constants are declared from the order of published values.
- Gap at the edge of the calibrated range in every search (0.8 mm).
- Rayleigh–Plesset is incompressible; peak pressures and focusing are upper bounds.
- Turbine, pump and motor efficiencies are declared.
- Next: CFD of the heater drums (OpenFOAM, Schnerr–Sauer and SST k-ω, as Sun et al. 2020), then a bench with KI dosimetry, hydrophones, a power analyser and a steam meter.

Run it

```
go run ./examples/harmonic_cavitation
go run ./examples/harmonic_cavitation -viz examples/harmonic_cavitation/viz
python examples/harmonic_cavitation/viz/plot_all.py
python examples/harmonic_cavitation/viz/make_report.py
go test ./examples/harmonic_cavitation
```

Sources

- L. Xue, Z. Hao, W. Ren et al., *Ultrason. Sonochem.* 109, 106999 (2024).
- X. Sun, X. Xuan, Y. Song et al., *Ultrason. Sonochem.* 70, 105311 (2020); X. Sun et al., *Ultrason. Sonochem.* (2021), doi:10.1016/j.ultsonch.2021.105771.
- M. Petkovšek, M. Zupanc, M. Dular et al., *Sep. Purif. Technol.* 118, 415 (2013).
- S. Kwon, J. Y. Yoon, *Proc. IMechE E* 227, 67 (2012), doi:10.1177/0954408912451535.

© 2026 fdamador@comcast.net

- Y.-H. Zhang et al., *Ultrason. Sonochem.* 124, 107727 (2025), review of rotational cavitators.
- Lord Rayleigh, *Phil. Mag.* 34, 94 (1917); C. E. Brennen, *Cavitation and Bubble Dynamics*, OUP 1995.
- A. B. Wood, *A Textbook of Sound*, 1930; H. Schlichting, *Boundary-Layer Theory*.
- D. White, P. Nylander, the Mandelbulb (2007–2009); D. Hvidtfeldt, distance-estimated 3-D fractals (2011).
- IAPWS-95 and IAPWS-IF97 (water); IAPWS R12-08 (viscosity).

© 2026 fdamador@comcast.net